

Product Guide

CAM-0402-BRG

Multiple Camera Switcher with A/V Bridge
(Bi-directional HDMI-USB)



WyreStorm®

**Multiple Camera Solution
Manufacturer**

Contents

Product Overview & Key Features

Why Choose Us?

Interface & Specifications

Typical Connection & Applications

Competitor Analysis

Quick Installation Guide & Web UI Introduction

Q & A Session



Powerful · Reliable · Simple

Product Overview

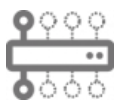


Multiple Camera Switcher with A/V Bridge (Bi-directional HDMI-USB)

The CAM-0402-BRG is a multi-camera switcher with USB and HDMI inputs (2x USB & 2x HDMI) and outputs (1x USB & 1xHDMI). It enables fast and seamless switching with no black screen, ensuring a smooth video experience. It also supports multiview modes (PIP, Pyramid, Quad, etc.), allowing multiple camera feeds to be displayed clearly within a single image – ideal for hybrid teaching and collaborative meetings. Additionally, the device functions as a bi-directional A/V bridge between HDMI and USB, allowing users to connect HDMI or USB cameras and route the signal to an HDMI monitor or laptop via the USB host port.

This highly versatile and easy-to-install solution is ideal for collaborative meetings and remote education scenarios where multiple camera angles are required, or HDMI content needs to be captured and transmitted along with camera feeds.

Main Features



Comprehensive Camera Input Options

Offers a variety of inputs, including USB and HDMI, ensuring compatibility with cameras used in conferences, meetings, and educational settings.



Fast Seamless Switching Between Multiple Cameras

Eliminate black screens during source switching between different cameras, enhancing viewing experience with smooth and clarity image.



Bi-directional A/V Bridge Between HDMI and USB

This advanced feature allows users to connect HDMI or USB cameras and stream the signal to an HDMI monitor or laptop via the USB host port—delivering a seamless and enhanced multi-camera viewing experience.



Flexible Control Approaches

Offers various control methods, such as control button, API (RS232 & Telnet), and Web-UI, allowing for convenient management and configuration of the camera.



Reliable and High-Quality Resolution

4K resolution guarantees sharp and clear video, delivering a fully immersive presentation experience.



Multiview Mode Support

Displays up to 4 camera sources simultaneously on one screen (PIP, Pyramid, Quad, etc.), to provide an efficient, comprehensive and real-time overview.



Independent 2-in-1 USB 3.0 Hub

The device integrates a 2-in-1 USB 3.0 hub, allowing connection of USB peripherals—like speakerphones or mice—to a laptop via the USB 3.0 host port, delivering streamlined integration and a smoother workspace experience.



Effortless Setup and User-friendly Operation

Plug-and-play with zero configuration required, offering hassle-free setup and immediate operation out of the box. This feature makes it exceptionally user-friendly, particularly in fast-paced or non-technical environments.

WyreStorm

Powerful · Reliable · Simple

Why Choose Us: 1. Immersive Viewing Experience

Deliver an immersive viewing experience with 4K@30Hz 4:4:4 resolution, seamless switching, and support for multiview modes – ensuring stunning detail and smooth visuals for the audience.

High-Quality Video Performance

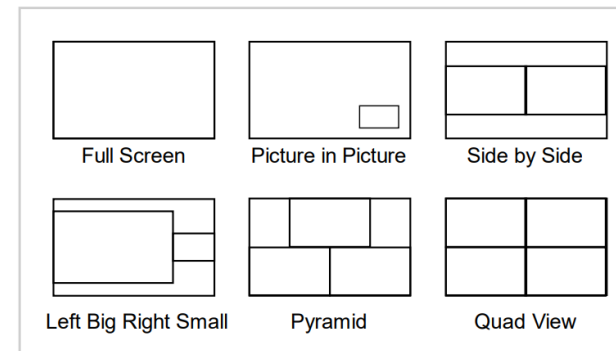
Provide high-quality 4K@30Hz 4:4:4 8bit resolution for detailed information on HDMI and USB output, providing superior visual quality for your presentations.

Fast and Seamless Source Switching

It delivers fast and seamless source switching without black screen, ensuring a consistently smooth video experience.

Multiview Mode Supports

Offers multiple layouts (PIP, Pyramid, Quad, etc.), to provide a comprehensive and real-time overview on a single screen – ideal for collaborative environments, monitoring, and streamlined decision-making.



WyreStorm®

Powerful · Reliable · Simple

Why Choose Us: 2. Creating Hybrid Environment with Multi-Camera

With excellent camera adaptability, Integrated 2-in-1 USB Hub, and efficient operation in hybrid environments, the system allows all meeting participants to easily contribute and collaborate productively.

Excellent Camera Adaptability

Equipped with a bi-directional A/V bridge (HDMI - USB), the device supports a diverse array of camera models. It enables the integration of multi-camera, intelligent video solution into the hybrid experience.

Built-in USB 3.0 Hub

Works flawlessly with extra speakerphones, keyboard, mouse, and other USB 3.0 peripherals, delivering a seamless and superior hybrid working experience.

Efficient Operation in Hybrid Meetings

Enhance the efficiency of hybrid meetings with a user-friendly Web-UI or by integrating with SYN-TOUCH10.



Powerful · Reliable · Simple

Why Choose Us: 3. Plug & Play, Guest-Friendly

Simplify setup with plug-and-play functionality, allowing for guests to connect and present without any hassle.

Flexible Control Methods

Offers multiple control methods (buttons, API, Web-UI), allowing for convenient management and configuration of the camera switcher.

Plug-and-Play, user-friendly

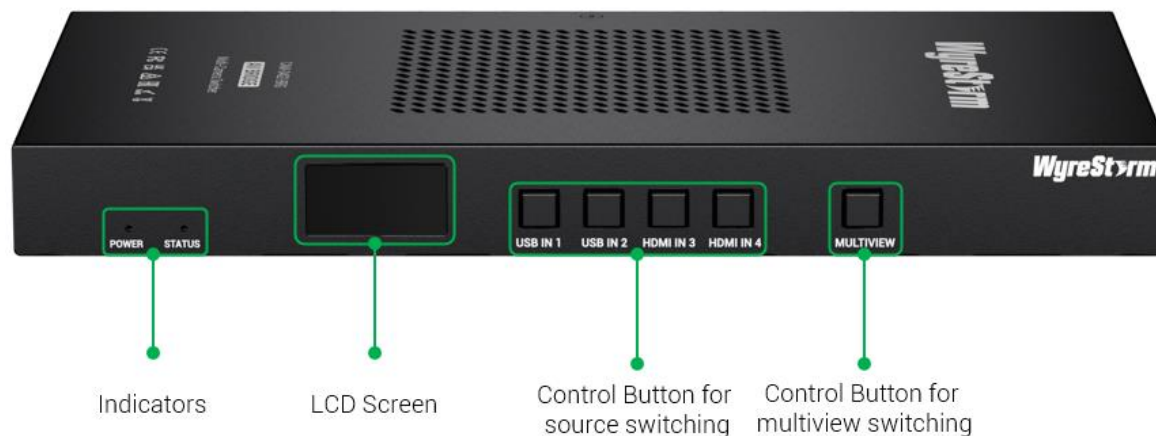
Plug-and-play functionality means zero configuration is required, offering hassle-free setup and immediate operation out of the box, making it exceptionally user-friendly.

Secure Network Access

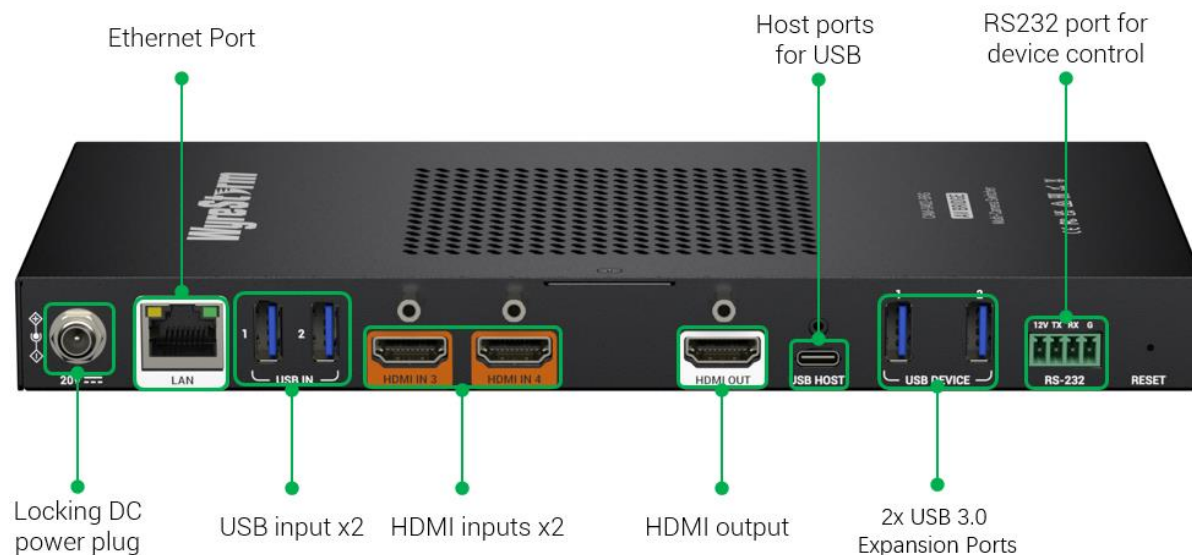
Independent Ethernet port, IP Setting and advanced protocols like HTTPs, Telnet over TLS, make sure the communication is security and user privacy is fully protected.

Interface of the Product

Front Panel



Rear Panel



Featuring:

2x USB IN & 2x HDMI IN

All four ports connection of 4K camera, and it offers fast seamless switching between different sources for a better viewing experience.

HDMI OUT & USB Host

Two output video signals, the USB Host output has mirroring HDMI output, both of these two outputs can support up to 4K@30Hz resolution.

2x USB3.0 Expansion Ports

Easily connect speakerphone, keyboard or other USB 3.0 peripherals to your laptop for flexible settings.

Specifications & Packing List

Main Specifications

Input Video Ports	2x USB 3.0, 2x HDMI (HDMI 1.4)
Output Video Ports	1x HDMI, 1x USB Host;
Video Resolution (Max)	HDMI Output: 4K@30Hz 4:4:4 8bit USB: 4K@30Hz (MJPEG)
Multiview Mode	PIP Side-by-Side Pyramid Quad etc.
USB	2x USB-A: USB 3.0 Max Bandwidth: 5.0 Gbps * USB peripherals connected to your PC
USB Protocols	UAC UVC
Ethernet	1x LAN: 8-pin RJ-45 Female Web-UI API
RS232	1x 4-pin 3.5mm Phoenix Connector (API Control)
Control Method	Front panel buttons Web-UI API (RS232 & Telnet)
Power Supply	20V DC 3A

What's in the Box?

- 1x CAM-0402-BRG;
- 1x 20V/3A Power Supply (US/UK/EU/AU);
- 1x 4-pin Phoenix Connector (for RS-232);
- 4x Mount Bracket and Screws;
- 1x Quick Start Guide;

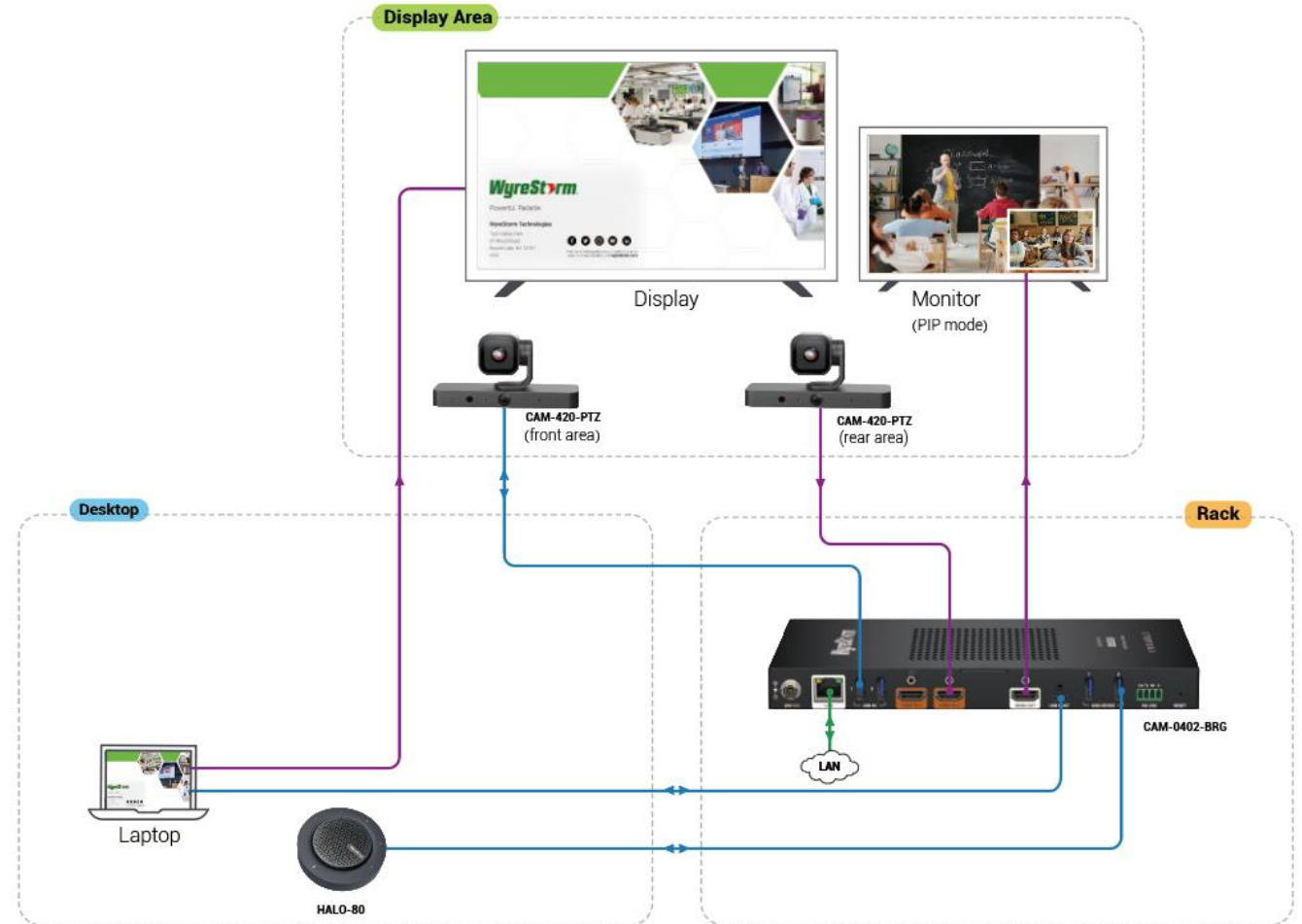
Typical Connection & Applications

Application Scenario 1: Conference Room, Classroom (Multi-camera)

This solution simplifies installation by supporting multiple cameras and featuring a plug-and-play design. It significantly improves the efficiency of remote meetings and educational sessions, ensuring every meeting is productive and seamless.

Key Benefits Include:

- **Fast and Seamless Multi-Camera Switching:** Switch between up to four HDMI/USB cameras with ultra-fast transitions and zero black screens—ensuring a fluid and uninterrupted viewing experience.
- **Bi-directional A/V Bridge between HDMI and USB:** A reliable HDMI - USB bi-directional bridge that is fully compatible with all USB and HDMI cameras.
- **Multiview Display Modes:** Supports multiple layouts (PIP, Pyramid, Quad, etc.) to provide a comprehensive, real-time overview on a single screen.
- **Flexible Room Configuration:** Provides multiple control methods, including Buttons, Web UI & API (Telnet, RS232), and a user-friendly Web UI.



Typical Connection & Applications

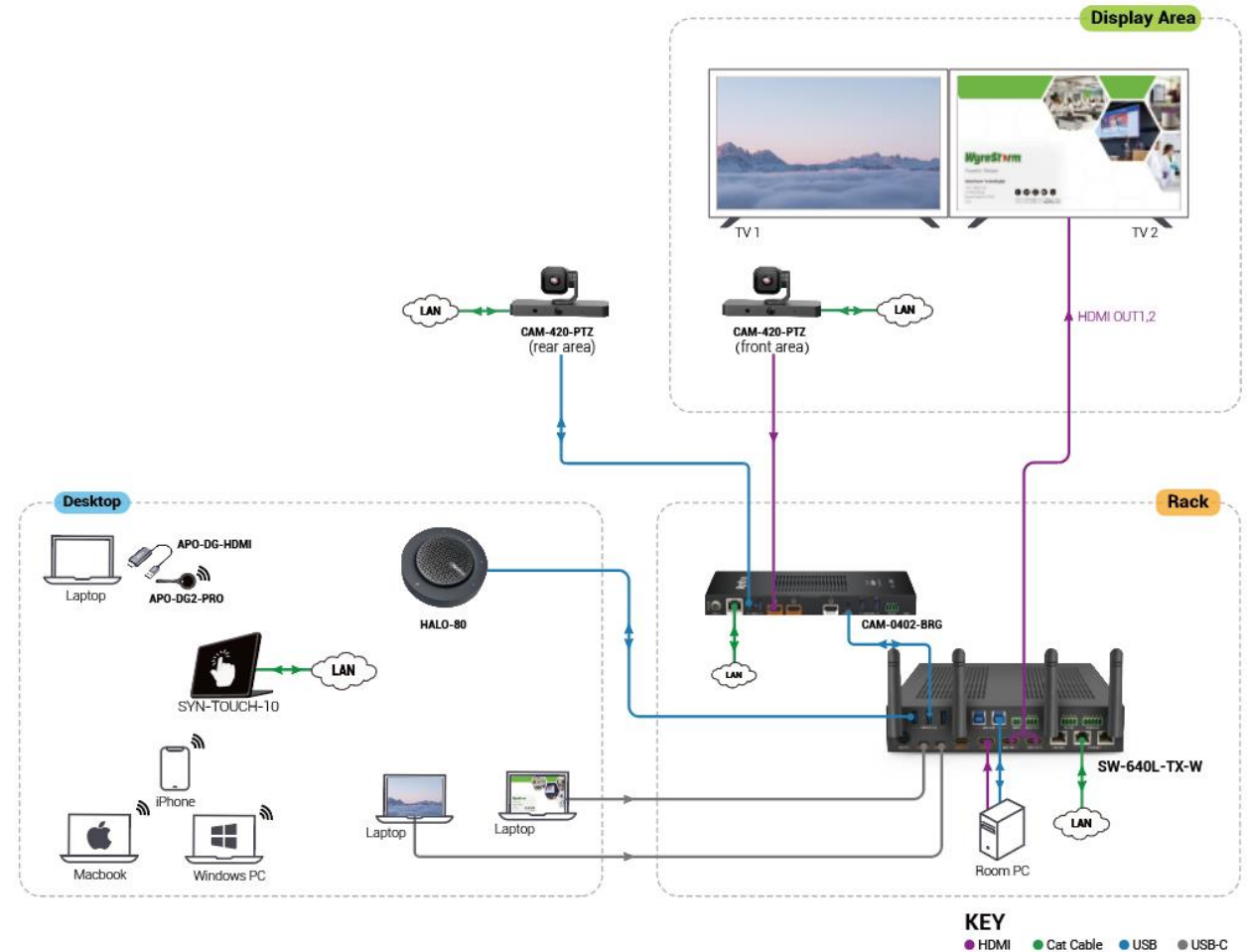
Application Scenario 2:

Multi-camera setups, working with other products

The WyreStorm multi-camera solution combines multiple cameras, an audio system, and flexible room configurations into a powerful setup. The camera switcher (CAM-0402-BRG) plays a critical role by handling multiple cameras, ensuring a fully immersive hybrid experience.

Key Benefits Include:

- **Better Wireless Casting Experience:** Working with SW-640L-TX-W, casting superior image by various cameras.
- **Fast and Seamless Multi-Camera Switching:** Switch between up to four HDMI/USB cameras with ultra-fast transitions and zero black screens—ensuring a fluid and uninterrupted viewing experience.
- **Multiview Display Modes:** Supports multiple layouts (PIP, Pyramid, Quad, etc.) to provide a comprehensive, real-time overview on a single screen.
- **Flexible Room Configuration:** Provides multiple control methods, including Buttons, Web UI & API (Telnet, RS232), and a user-friendly Web UI.



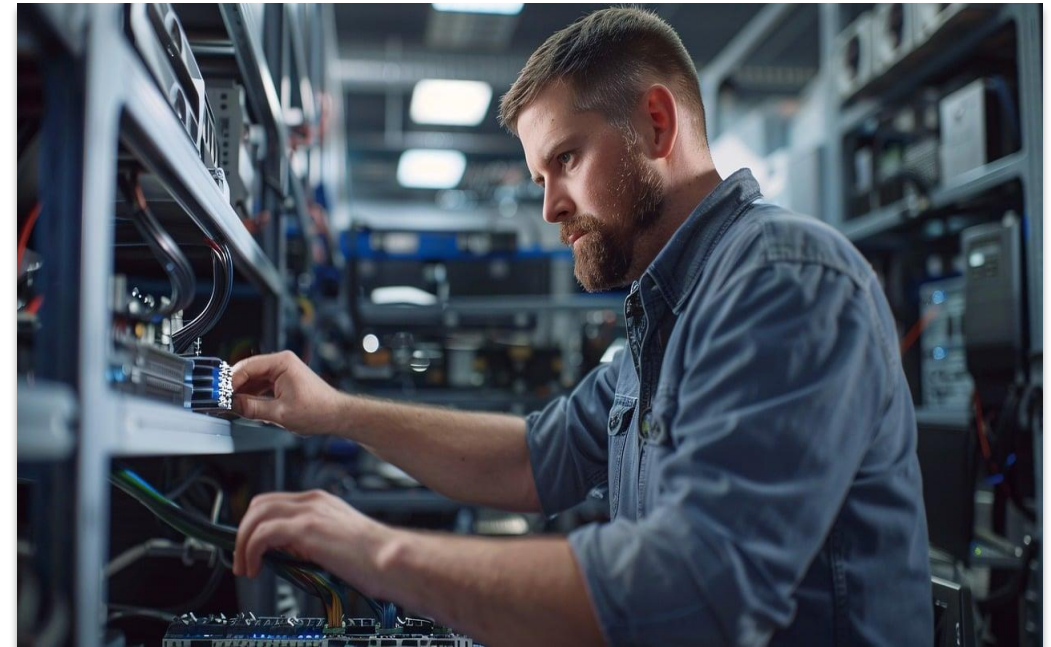
Comparison with Other Competitors

Specification		CAM-0402-BRG	InfoBit iTrans UC-HUB2U	AVer MT300N	Inogeni Share2U
Input	Video Input	2x HDMI	1x HDMI	3x HDMI	1x HDMI
	Video Input Timing	4K@30Hz 4:4:4 8bit	1080p@60Hz	1080p@60Hz	1080p@60Hz
	USB Input	2x USB 3.0 Type-A	1x USB 2.0 Type-A 1x USB 3.0 Type-A	3x USB 2.0 Type-A	2x USB 2.0 Type-A
	USB Input Timing	1x 4K@30Hz (or 2x 1080P@30Hz)	1080p@30Hz	1080p@60Hz	1080p@30Hz
Output	Video Output	1x HDMI	1x HDMI	2x HDMI	1x HDMI
	Video Output Timing	4K@30Hz 4:4:4 8bit	4K@60Hz	4K@60Hz	1080p@60Hz
	USB Output	1x USB 3.0 Type-C	1x USB 3.0 Type-C	2x USB 3.0 Type-B	1x USB 3.0 Type-B
	USB Output Timing	4K@30Hz	1080P@30Hz	Unknown	1080p@60Hz
	Multiview	Quad View (Up to 4 windows)	Dual View	No	Dual View
Others	USB Peripheral Interface	2x USB 3.0 Type-A	No	No	2x USB 3.0 Type-A
	Audio Interface	No	1x Analog in, 3.5mm TRS Jack 1x Analog Out, 3.5mm TRS Jack	Dante USB	1x Analog in, 3.5mm TRS Jack 1x Analog Out, 3.5mm TRS Jack
	RS232	Yes	Yes	No	Yes
	Control Approach	Button Web API (RS232/Telnet)	Button Web API (RS232/Telnet)	Automatic (Active Position) Web API (RS422)	USB HID Button API (RS232/Telnet)
	Additional Feature	Audio mixer	Audio mixer PoE		Audio mixer

Quick Installation Guide

Effortlessly set up your camera switcher with simple steps, ensuring seamless configuration and an exceptional hybrid experience.

- **Connect Interfaces:** Follow the standard product connection guidelines and customize connections based on your environment's layout for optimal performance.
- **Start the Device:** Use 'Quick Start Guide' to locate the device's IP address, then log into the Web-UI for easy equipment configuration.
- **Select the mode of image:** Select your multiview layout using the Web-UI. For detailed instructions, refer to the user manual.
- **Enjoy Superior Visuals:** Experience stunning video clarity and immersive visuals tailored to your needs.



Important Notes for Installation

- Ensure all cables are securely connected to avoid performance issues.
- Check your network settings and confirm that the device is on the correct subnet.
- Double-check configuration settings in the Web-UI to optimize display and functionality.
- For complex installations, refer to the documentation or seek professional assistance if necessary.

Introduction of Web Configuration Interface

The Web UI of the CAM-0402-BRG camera switcher is designed for intuitive device management and control through any web browser. With the Web UI, you can easily configure and select multiview mode with following features:

Web UI Functionality:

- **Interface:** Effortlessly manage device settings via a web browser.
- **Setting up:** Camera source routing, audio settings, multiview mode select, and USB settings.
- **Control:** Set up and adjust camera source inputs, outputs, and system settings.

Key Web UI Configurations:

- **General Settings:** Configure IP settings, manage security, and perform firmware updates.
- **Video Inputs/Outputs:** Manage camera source, select HDMI or USB source.
- **Multiview Settings:** Easy setting up to select the multiview mode.
- **Control Settings:** Enable RS-232 for device control and CEC for display control.

Final Thoughts

Q & A Session

Got questions? Don't hesitate to ask away.

